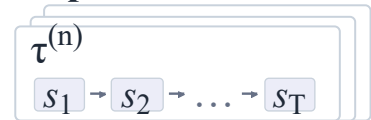
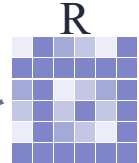


Spectral reference space

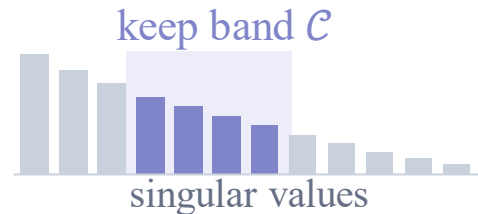
Step-unlabeled failure trajectories $\mathcal{D}_{\text{fail}}$



Frozen LLM



$$\text{SVD} \\ R = U\Sigma V^T$$

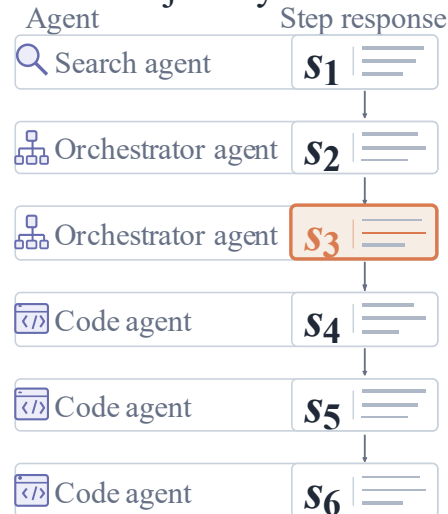


Fixed spectral subspace

$$\text{span}\{V_{:,c} : c \in \mathcal{C}\}$$

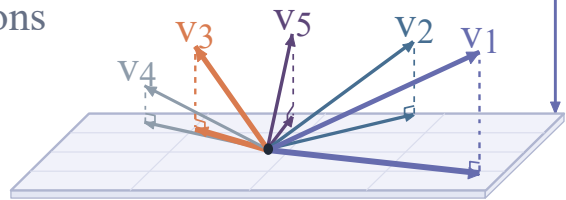
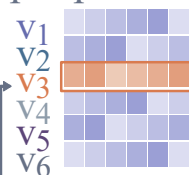
Failure attribution

Failed trajectory τ

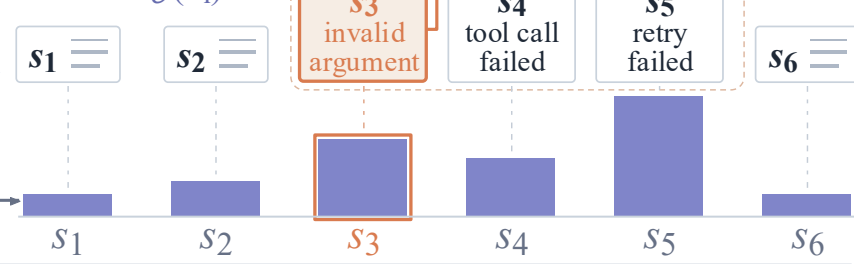


(2) Spectral base score

Step representations

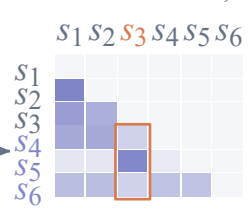


$$\text{Base score}_1 \\ S(s_i) = \frac{1}{\pi_{\mathcal{C}}(v_i) + \epsilon}$$



(3) Attention-guided rescoring

Attention $w_{i,t}$



$$B_i = \frac{\sum_{t>i} w_{i,t} S(s_t)}{\sum_{t>i} w_{i,t}}$$



Rescored score

